

Appendix 2: Per-policy Online Behavior

This supplementary appendix complements the online evaluation section of the main manuscript by reporting detailed per-policy execution traces and monitoring screenshots collected during the Day 2 heavy read workload with `WeightedWordCount`. Five executions were performed for each policy in the main evaluation. To avoid duplicating a large number of nearly identical monitoring figures, this appendix shows the screenshots associated with one selected run per policy, corresponding to the best observed completion time available for that policy. These screenshots are provided only to document the execution profile, HDFS state, and resource behavior of the selected run. All quantitative comparisons in the main manuscript are based on the five-run mean, sample standard deviation, and 95% confidence interval rather than on the screenshots shown here.

Interpretation of the screenshots. The figures below should therefore be read as qualitative execution evidence for the selected best observed run of each policy. They are not intended to replace the repeated-run statistics reported in the main manuscript.

HDFS default (RF = 3):

Under the default HDFS policy all blocks of the Day2 hot set are stored with a fixed replication factor of three and placed by the standard rack aware placement algorithm. The `fsck` report on `/exp/day2/data` confirms that the directory contains exactly 10 000 single block files of about 64 MiB each, with no missing or under replicated blocks. The NameNode report (Figure 1) shows that these 10 000 blocks are spread quite evenly across the ten DataNodes, with roughly 3 000 blocks and about 190 GB of HDFS data per node and only a small deviation between the lightest and the heaviest node. This layout is therefore a reasonable baseline for a balanced replicated configuration.

Node	Http Address	Last contact	Last Block Report	Used	Non DFS Used	Capacity	Blocks	Block pool used	Version
✓/default-rack/vmhd-dn02:9866 (172.23.10.152:9866)	http://vmhd-dn02:9866	1s	192m	189.25 GB	64.11 MB	294.29 GB	3044	189.25 GB (64.31%)	3.3.6
✓/default-rack/vmhd-dn05:9866 (172.23.10.155:9866)	http://vmhd-dn05:9866	2s	96m	189.26 GB	64.11 MB	294.29 GB	3058	189.26 GB (64.31%)	3.3.6
✓/default-rack/vmhd-dn09:9866 (172.23.10.159:9866)	http://vmhd-dn09:9866	1s	347m	194.58 GB	64.1 MB	294.29 GB	3135	194.58 GB (66.12%)	3.3.6
✓/default-rack/vmhd-dn01:9866 (172.23.10.151:9866)	http://vmhd-dn01:9866	1s	83m	189.36 GB	4.65 GB	294.29 GB	3055	189.36 GB (64.34%)	3.3.6
✓/default-rack/vmhd-dn06:9866 (172.23.10.156:9866)	http://vmhd-dn06:9866	2s	227m	189.25 GB	64.12 MB	294.29 GB	3045	189.25 GB (64.31%)	3.3.6
✓/default-rack/vmhd-dn10:9866 (172.23.10.160:9866)	http://vmhd-dn10:9866	1s	71m	189.26 GB	64.13 MB	294.29 GB	3048	189.26 GB (64.31%)	3.3.6
✓/default-rack/vmhd-dn04:9866 (172.23.10.154:9866)	http://vmhd-dn04:9866	1s	289m	203.56 GB	64.11 MB	294.29 GB	3281	203.56 GB (69.17%)	3.3.6
✓/default-rack/vmhd-dn08:9866 (172.23.10.158:9866)	http://vmhd-dn08:9866	1s	272m	180.37 GB	64.13 MB	294.29 GB	2902	180.37 GB (61.29%)	3.3.6
✓/default-rack/vmhd-dn07:9866 (172.23.10.157:9866)	http://vmhd-dn07:9866	2s	24m	188.88 GB	5.1 GB	294.29 GB	3050	188.88 GB (64.18%)	3.3.6
✓/default-rack/vmhd-dn03:9866 (172.23.10.153:9866)	http://vmhd-dn03:9866	1s	75m	176.41 GB	64.13 MB	294.29 GB	2850	176.41 GB (59.94%)	3.3.6

Figure 1: Per node HDFS usage after the Default replication plan has been applied.

In the selected Default run shown in this appendix, the Day2 access pattern is replayed with **WeightedWordCount**. The job reads about 6.7 TiB of data from HDFS for the 99 883 logical reads encoded in **weights.csv**, with 201 successful map tasks. The JobHistory view in Figure 2 shows a classic map only profile, with most maps finishing in the same time window and a small number of slower attempts that are rescheduled on other nodes. There is no reduce phase, no failed task, and no read from erasure coded blocks in this run.

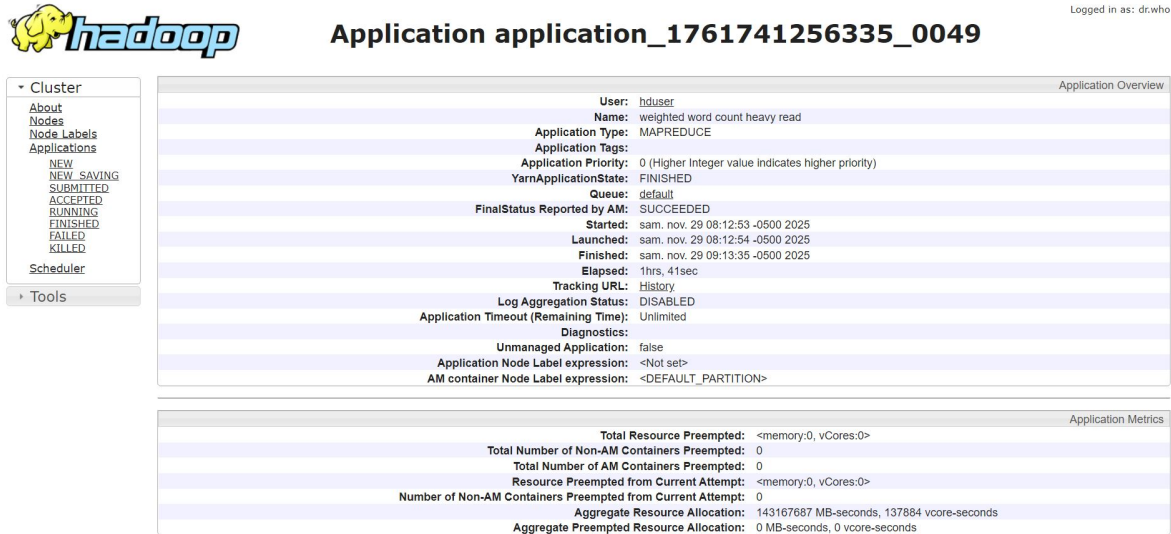


Figure 2: YARN application summary for the Default **WeightedWordCount** job.

The Ganglia panels for the Default run confirm that the load is mainly I/O bound. At the cluster level, the overview in Figure 3 shows a sustained CPU utilization around 35 to 45 % during the main part of the job, with a visible **wa** component and a large idle fraction, which indicates that the workload is limited more by disk and network than by computation. Per node graphs in Figure 7 report user time around 3 % and system time around 5 to 7 % on each DataNode, while the NameNode stays at a moderate user utilization close to 30 %. The aggregated network panel in Figure 5 shows a clear plateau

between 14:20 and 14:40 where the cluster receives and sends close to 3 GB/s, and the packet level views in Figure 6 confirm that this traffic is well shared across nodes. Finally, memory usage remains stable on all machines (Figure 4), with about 10 to 11 GB used per DataNode and no swap activity. This Default run therefore acts as a healthy reference point with a balanced replicated layout, moderate CPU pressure, and a network and disk footprint that match the heavy read workload imposed by WeightedWordCount.

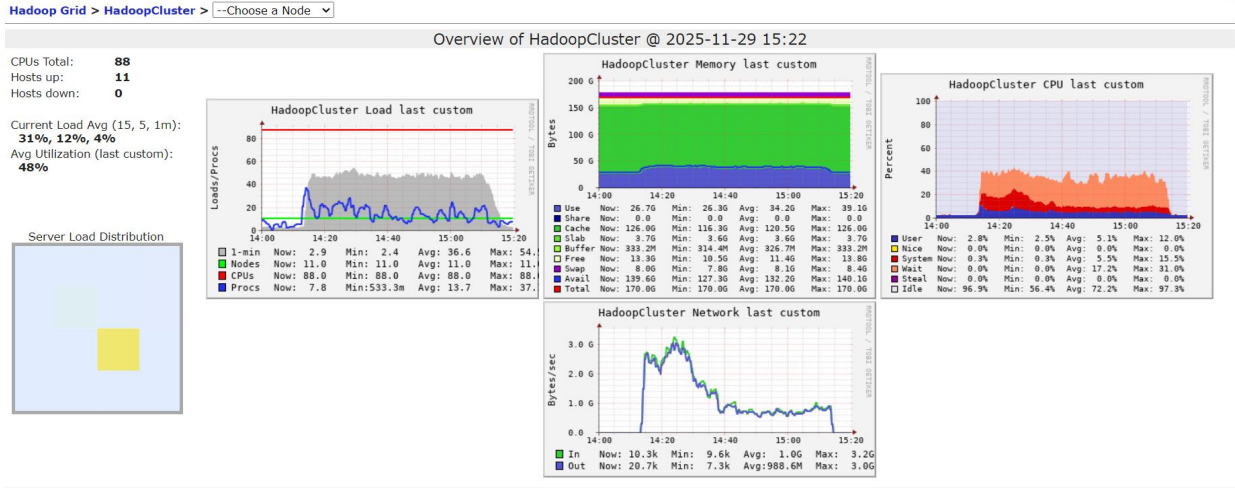


Figure 3: Ganglia overview of the Hadoop cluster during the Default WeightedWordCount run.



Figure 4: Per node memory usage during the Default run.

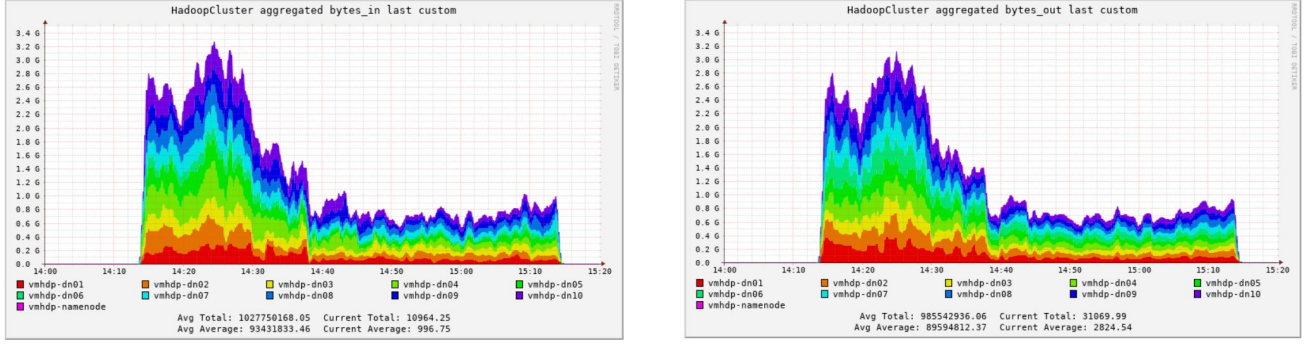


Figure 5: Aggregated incoming (left) and outgoing (right) network throughput for the Default run.

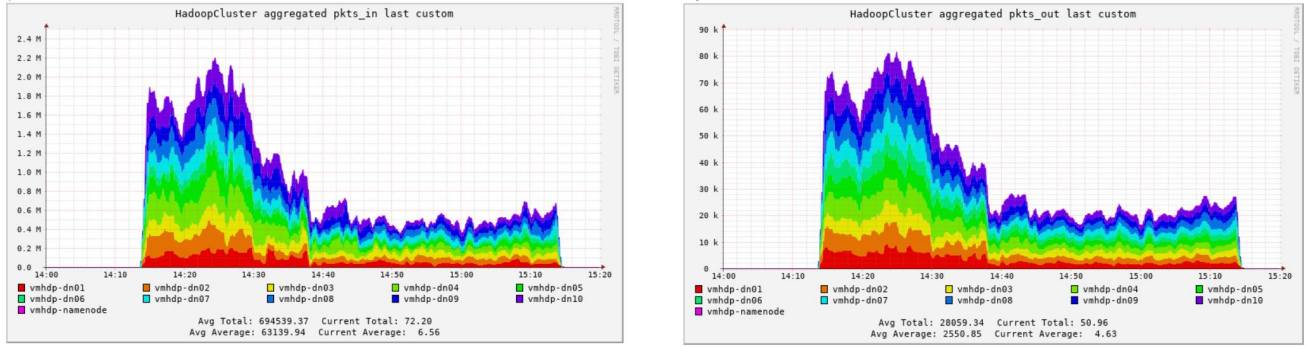


Figure 6: Aggregated incoming and outgoing packets per second for the Default run.

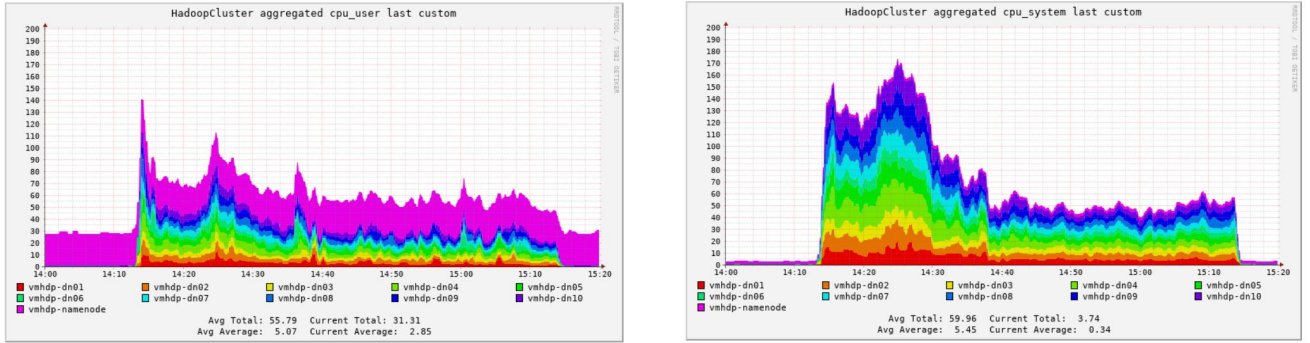


Figure 7: Per node user and system CPU utilization during the Default run.

DRPMLC:

Before running the `WeightedWordCount` workload under DRPMLC, we checked the consistency of the hot dataset. The `fsck` report on `/exp/day2/data` confirms that the 10 000 files are still stored as standard replicated HDFS blocks with a single block per file, an average replication factor of three, and no missing, corrupt, over or under replicated blocks. Erasure coded block groups are reported as zero for this path, which means that the DRPMLC plan keeps the hot subset fully replicated, in line with its original design for cold data.

The HDFS capacity report taken after the DRPMLC run shows that disk usage remains well balanced across the ten DataNodes. Individual DFS utilization varies between about 60 % and 68 %, with per node used space between roughly 176 GB and 202 GB. The total DFS used is close to 1.85 TB out of

2.72 TB of present capacity, very close to the Default layout. DRPMLC therefore does not introduce noticeable storage skew on this cluster.

Node	Http Address	Last contact	Last Block Report	Used	Non DFS Used	Capacity	Blocks	Block pool used	Version
✓/default-rack/vmhdn-dn02:9866 (172.23.10.152:9866)	http://vmhdn-dn02:9864	2s	212m	176.03 GB	64.12 MB	294.29 GB	2840	176.03 GB (59.81%)	3.3.6
✓/default-rack/vmhdn-dn05:9866 (172.23.10.155:9866)	http://vmhdn-dn05:9864	1s	116m	189.01 GB	64.12 MB	294.29 GB	3050	189.01 GB (64.22%)	3.3.6
✓/default-rack/vmhdn-dn09:9866 (172.23.10.159:9866)	http://vmhdn-dn09:9864	2s	6m	198.3 GB	64.12 MB	294.29 GB	3203	198.3 GB (67.38%)	3.3.6
✓/default-rack/vmhdn-dn01:9866 (172.23.10.151:9866)	http://vmhdn-dn01:9864	0s	103m	181.1 GB	4.65 GB	294.29 GB	2928	181.1 GB (61.54%)	3.3.6
✓/default-rack/vmhdn-dn06:9866 (172.23.10.156:9866)	http://vmhdn-dn06:9864	1s	246m	178.04 GB	64.12 MB	294.29 GB	2875	178.04 GB (60.5%)	3.3.6
✓/default-rack/vmhdn-dn10:9866 (172.23.10.160:9866)	http://vmhdn-dn10:9864	2s	91m	202.24 GB	64.14 MB	294.29 GB	3260	202.24 GB (68.72%)	3.3.6
✓/default-rack/vmhdn-dn04:9866 (172.23.10.154:9866)	http://vmhdn-dn04:9864	2s	308m	198.08 GB	64.12 MB	294.29 GB	3199	198.08 GB (67.31%)	3.3.6
✓/default-rack/vmhdn-dn08:9866 (172.23.10.158:9866)	http://vmhdn-dn08:9864	1s	291m	182.64 GB	64.13 MB	294.29 GB	2949	182.64 GB (62.06%)	3.3.6
✓/default-rack/vmhdn-dn07:9866 (172.23.10.157:9866)	http://vmhdn-dn07:9864	1s	44m	191.65 GB	5.1 GB	294.29 GB	3096	191.65 GB (65.12%)	3.3.6
✓/default-rack/vmhdn-dn03:9866 (172.23.10.153:9866)	http://vmhdn-dn03:9864	2s	94m	193.04 GB	64.13 MB	294.29 GB	3120	193.04 GB (65.59%)	3.3.6

Figure 8: Per node HDFS usage after the DRPMLC replication plan has been applied.

The job history of the selected DRPMLC run shows that the heavy **WeightedWordCount** job processes the same logical workload as in the Default case. The custom counter reports 6.70×10^{12} bytes read from HDFS and the input format still sees 10 001 map input records, that is exactly one record per line in **weights.csv**. The job launches 206 map tasks, with 201 successful attempts and 5 killed attempts. In the displayed run, the job runs from 00:36:12 to 01:59:29, for a total of 1 hour, 23 minutes and 17 seconds. The average time per map task is 14 minutes 55 seconds, and several maps run for more than 25 minutes. The repeated-run comparison used in the main manuscript is reported separately from this selected execution trace.

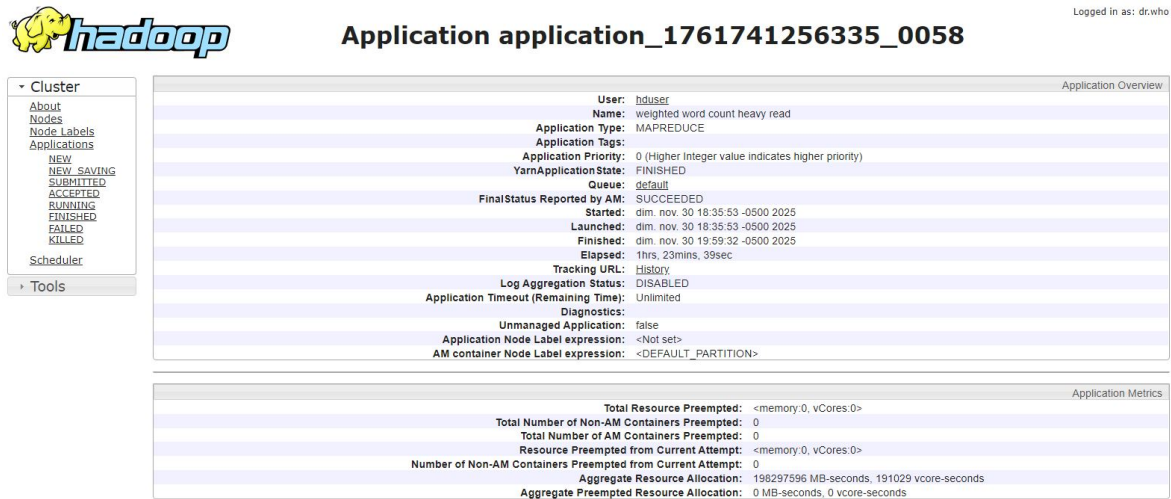


Figure 9: YARN application summary for the DRPMLC **WeightedWordCount** job.

Cluster wide Ganglia metrics confirm that the cluster remains busy for a longer period under DRPMLC. The overview panel in Figure 10 shows a first ramp up phase around 00:40, followed by a

sustained load plateau where the one minute load hovers between 35 and 50 runnable processes. During this phase the aggregate network throughput (bottom of the overview) reaches between 2 and 2.5 GB/s, comparable to the Default run, but the period of high activity extends further in time. The CPU panel indicates that the total user plus system utilization of the 88 vCPUs remains mostly in the 30 % to 45 % range for more than one hour, again a bit longer than with the Default placement.

The per node HDFS view in Figure 8 confirms that DRPMLC does not create pathological hotspots. The heaviest DataNode stores around 202 GB while the lightest one is around 176 GB, which corresponds to a spread of less than 12 percentage points in DFS utilization. Combined with the healthy `fsck` report, this indicates that the longer runtime is not due to storage imbalance but to the way replicas are placed relative to the compute tasks.

From a resource usage perspective, memory remains stable throughout the run. Figure 11 shows that each DataNode uses roughly 10 to 12 GB of RAM, with only minor variations over time and no sign of swapping. The cluster overview gives a similar picture, with total used memory around 120 GB out of 170 GB. This suggests that the workload is I/O bound rather than memory bound.

Network level graphs in Figures 12 and 13 provide more insight into the cost of the DRPMLC placement. Aggregated incoming throughput peaks close to 2.8 GB/s and outgoing throughput close to 3.0 GB/s during the main phase of the run. Compared to Default, the peak values are of the same order of magnitude, but the high throughput period lasts longer and the tail after 01:10 shows persistent medium level traffic while some late map tasks finish. The packets per second plots follow the same pattern, with a dense burst of traffic in the first half of the experiment and a longer low to medium intensity tail.

Finally, CPU utilization per node, detailed in Figure 14, shows that DRPMLC uses the DataNodes quite uniformly. User and system CPU stacks display the same long plateau as in the cluster overview, with moderate but steady activity and no obvious outliers among the nodes. Together with the unchanged logical workload, these observations indicate that the higher completion time of DRPMLC is mainly due to longer data paths and reduced locality for some map tasks rather than to a lack of resources or to storage imbalance.

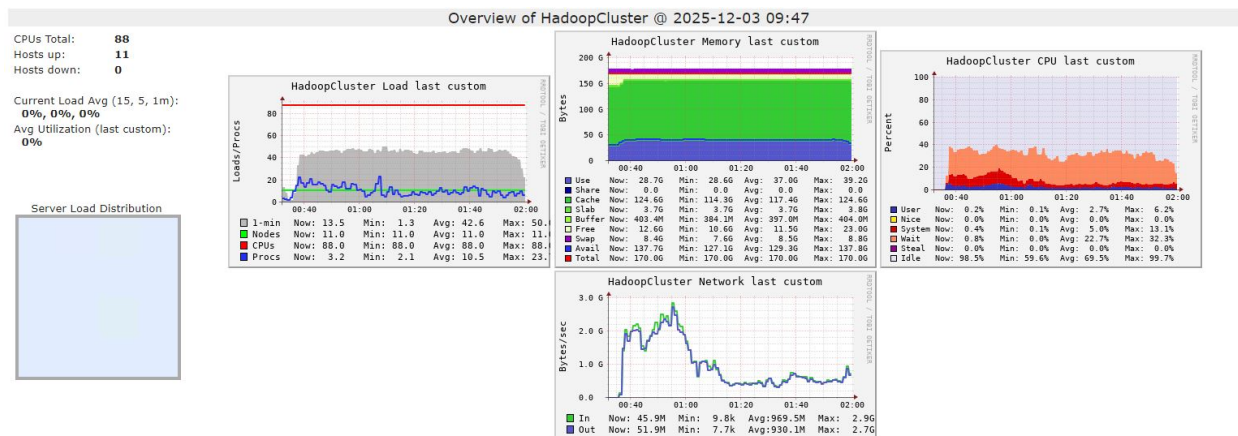


Figure 10: Ganglia overview of the Hadoop cluster during the DRPMLC WeightedWordCount run.



Figure 11: Per node memory usage during the DRPMLC run.

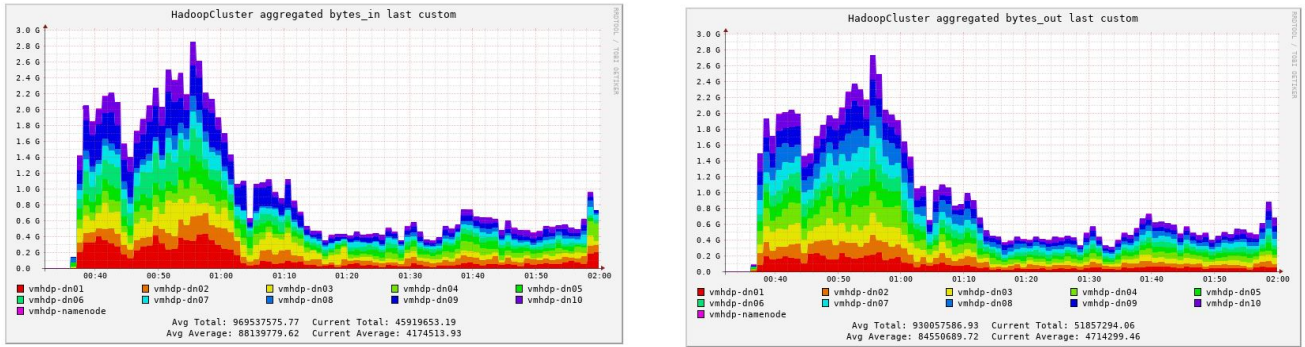


Figure 12: Aggregated incoming (left) and outgoing (right) network throughput for the DRPMLC run.

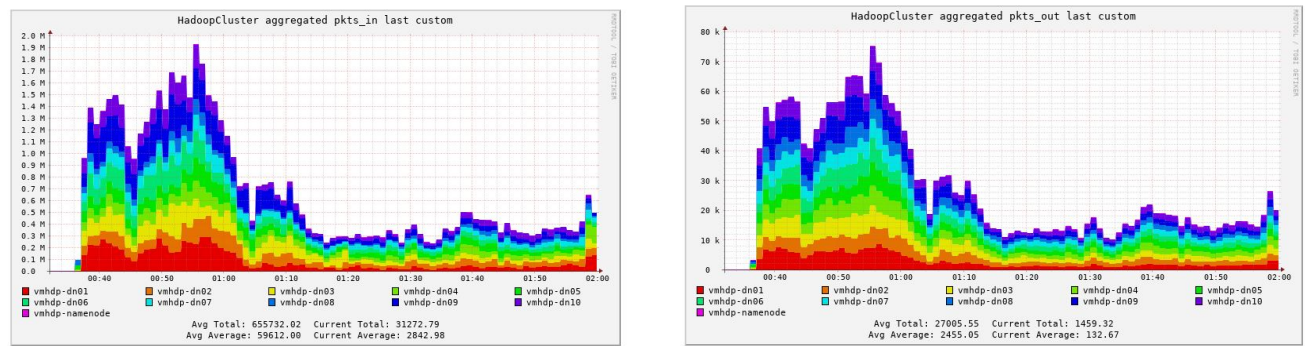


Figure 13: Aggregated incoming and outgoing packets per second for the DRPMLC run.

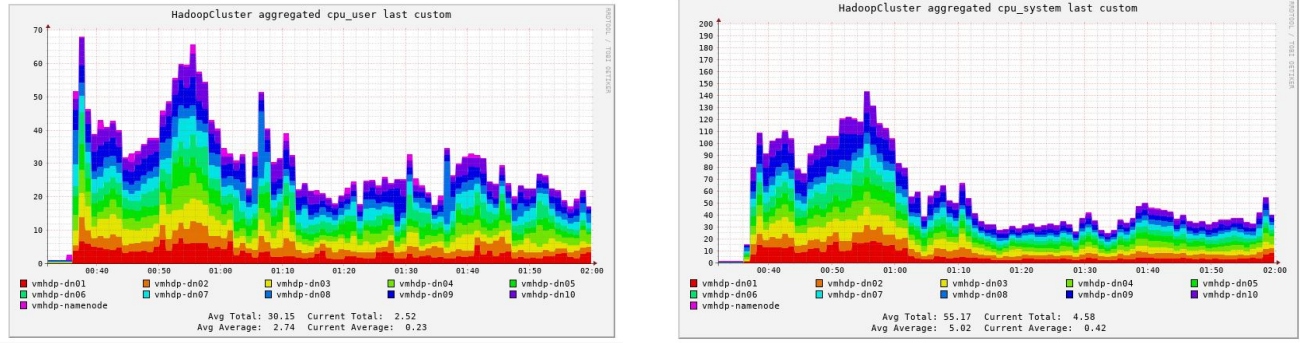


Figure 14: Per node user and system CPU utilization during the DRPMLC run.

ERMS:

Under the ERMS layout, the heavy `WeightedWordCount` job keeps exactly the same logical workload as in the other policies. The selected run shown here reports an elapsed time of 1 h 2 min 12 s. Map input and output counters confirm that the run processes 6.7 TB of compressed HDFS input and shuffles about 794 MB of intermediate data. The repeated-run timing comparison is reported in the main manuscript; the purpose of this appendix is to document the execution and resource profile of the selected run.

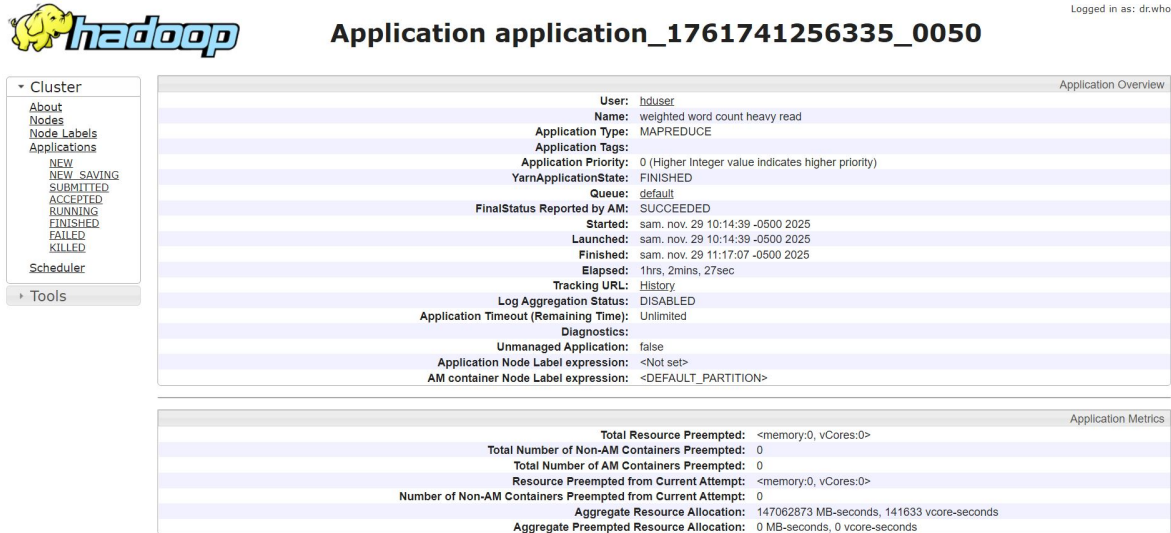


Figure 15: YARN application summary for the ERMS `WeightedWordCount` job.

From the HDFS report, ERMS pushes the system much closer to its storage limit. The NameNode reports 2.46 TB of DFS used space, which corresponds to about 90.6% of the available HDFS capacity, compared to roughly 68% in the Default run. The DataNode view shows that each node stores between 246 GB and 257 GB, with an average around 253 GB per node, that is about 84% to 87% of the 294 GB capacity. The distribution is very even across the ten DataNodes, which indicates that ERMS preserves good balance while increasing the overall replication factor for popular files.

Node	Http Address	Last contact	Last Block Report	Used	Non DFS Used	Capacity	Blocks	Block pool used	Version	
✓/default-rack/vmhdp-dn02:9866 (172.23.10.152:9866)	http://vmhdp-dn02:9864	2s	281m	251.88 GB	64.12 MB	294.29 GB		4042	251.88 GB (85.59%)	3.3.6
✓/default-rack/vmhdp-dn05:9866 (172.23.10.155:9866)	http://vmhdp-dn05:9864	1s	185m	254.65 GB	64.12 MB	294.29 GB		4098	254.65 GB (86.53%)	3.3.6
✓/default-rack/vmhdp-dn09:9866 (172.23.10.159:9866)	http://vmhdp-dn09:9864	1s	76m	250.28 GB	64.12 MB	294.29 GB		4026	250.28 GB (85.05%)	3.3.6
✓/default-rack/vmhdp-dn01:9866 (172.23.10.151:9866)	http://vmhdp-dn01:9864	0s	172m	246.87 GB	4.65 GB	294.29 GB		3973	246.87 GB (83.89%)	3.3.6
✓/default-rack/vmhdp-dn06:9866 (172.23.10.156:9866)	http://vmhdp-dn06:9864	1s	315m	256.98 GB	64.12 MB	294.29 GB		4126	256.98 GB (87.32%)	3.3.6
✓/default-rack/vmhdp-dn10:9866 (172.23.10.160:9866)	http://vmhdp-dn10:9864	2s	160m	252.92 GB	64.14 MB	294.29 GB		4064	252.92 GB (85.94%)	3.3.6
✓/default-rack/vmhdp-dn04:9866 (172.23.10.154:9866)	http://vmhdp-dn04:9864	1s	17m	252.33 GB	64.12 MB	294.29 GB		4064	252.33 GB (85.74%)	3.3.6
✓/default-rack/vmhdp-dn08:9866 (172.23.10.158:9866)	http://vmhdp-dn08:9864	0s	1m	253.77 GB	64.13 MB	294.29 GB		4074	253.77 GB (86.23%)	3.3.6
✓/default-rack/vmhdp-dn07:9866 (172.23.10.157:9866)	http://vmhdp-dn07:9864	0s	113m	246.87 GB	5.1 GB	294.29 GB		3975	246.87 GB (83.89%)	3.3.6
✓/default-rack/vmhdp-dn03:9866 (172.23.10.153:9866)	http://vmhdp-dn03:9864	0s	164m	253.64 GB	64.13 MB	294.29 GB		4080	253.64 GB (86.19%)	3.3.6

Figure 16: Per node HDFS usage after the ERMS replication plan has been applied.

The Ganglia overview in Figure 17 shows a network intensive phase at the beginning of the run, followed by a long plateau at lower throughput. The aggregated bytes in and bytes out curves in Figure 19 peak around 2 to 2.5 GB/s during the first 20 minutes, then settle to a sustained level below 1 GB/s for the remainder of the job. The packet rate in Figure 20 follows the same pattern, with outgoing traffic reaching several tens of thousands of packets per second and incoming traffic peaking close to 1.5 to 1.8 million packets per second at cluster level before dropping to a moderate steady state. This behavior is consistent with ERMS creating and placing additional replicas early in the run, then benefiting from improved locality during the long tail of repeated reads.

CPU and memory metrics confirm that ERMS stresses storage more than compute. The aggregated user and system CPU graphs in Figure 21 show moderate utilization on the DataNodes, with short spikes when containers start or when many replicas are created, but average CPU usage remains well below saturation on each node. The per node memory dashboard in Figure 18 indicates that all DataNodes keep several gigabytes of free memory and a large buffer cache, with no visible swapping. Overall, ERMS trades a significant increase in HDFS space consumption for better read locality and a shorter completion time than DRPMLC, while keeping CPU and memory overhead at a comfortable level.

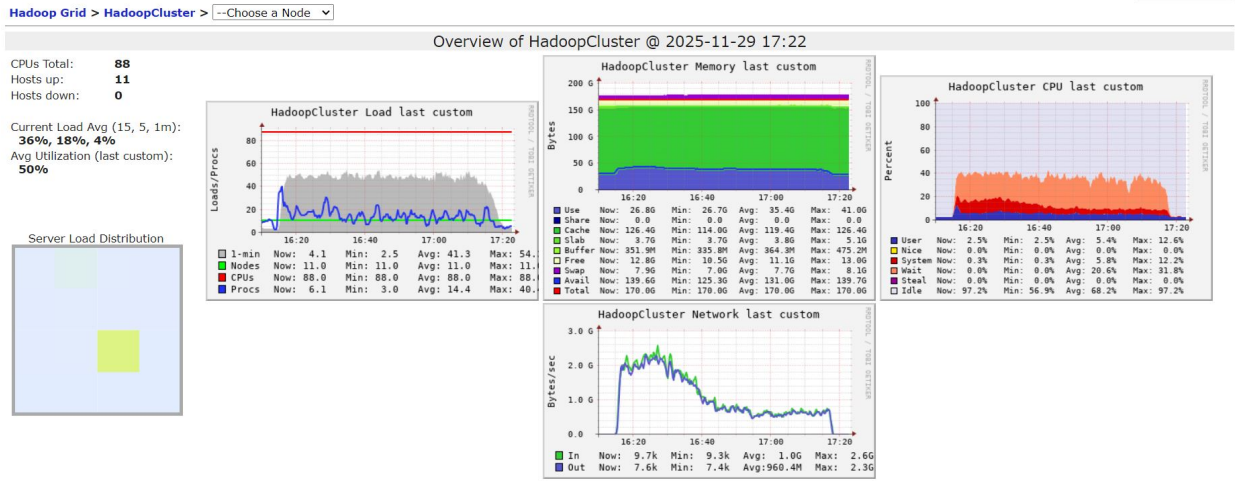


Figure 17: Ganglia overview of the Hadoop cluster during the ERMS WeightedWordCount run.



Figure 18: Per node memory usage during the ERMS run.

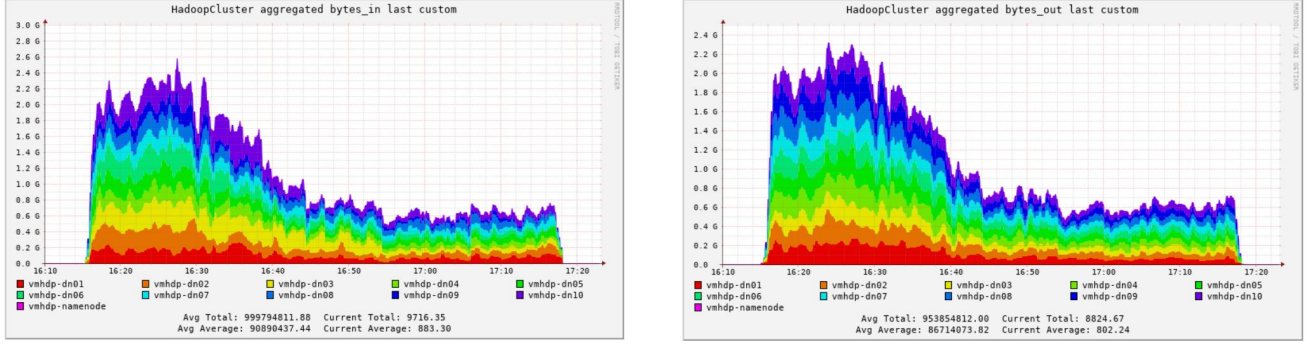


Figure 19: Aggregated incoming (left) and outgoing (right) network throughput for the ERMS run.

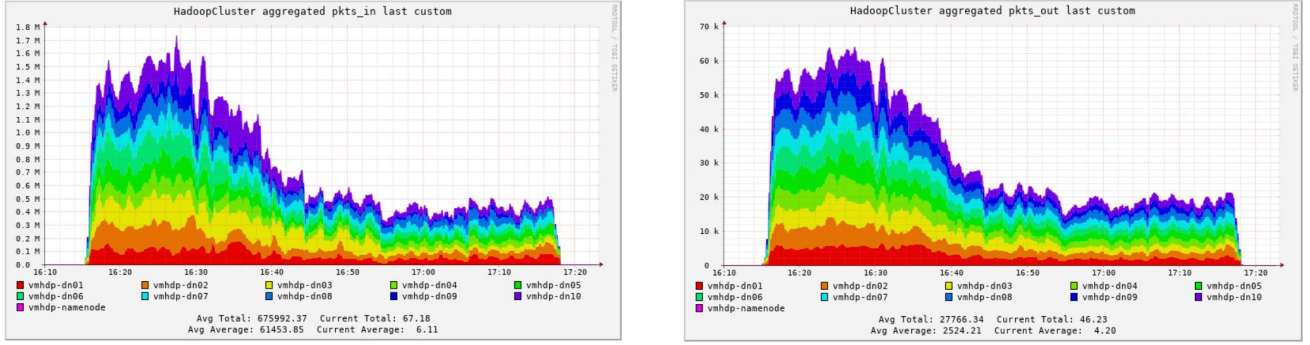


Figure 20: Aggregated incoming and outgoing packets per second for the ERMS run.

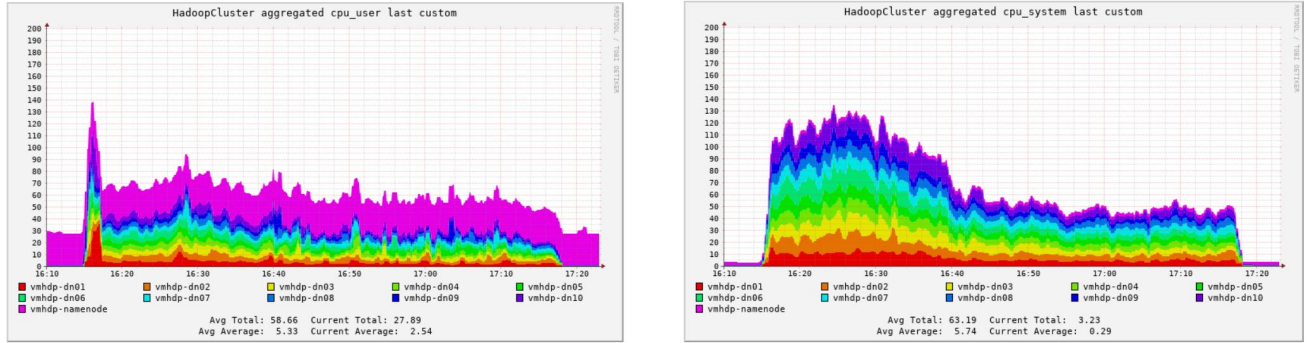


Figure 21: Per node user and system CPU utilization during the ERMS run.

SBR:

Under the SBR layout, the heavy **WeightedWordCount** job keeps exactly the same logical workload as in the other policies. The YARN history for the selected run reports an elapsed time of about 58 min (Figure 22). The job uses 203 map tasks, of which 201 succeed on the first attempt and 2 are retried, and processes 10 000 map input records corresponding to the 10K most frequently accessed files in the Day 2 workload. Custom counters confirm that the job issues about 6.7 TB of HDFS reads and shuffles a similar amount of intermediate data as in the other runs. The main manuscript uses the statistics from all five executions rather than this individual run for policy comparison.



Application application_1761741256335_0056

Logged in as: dr:who

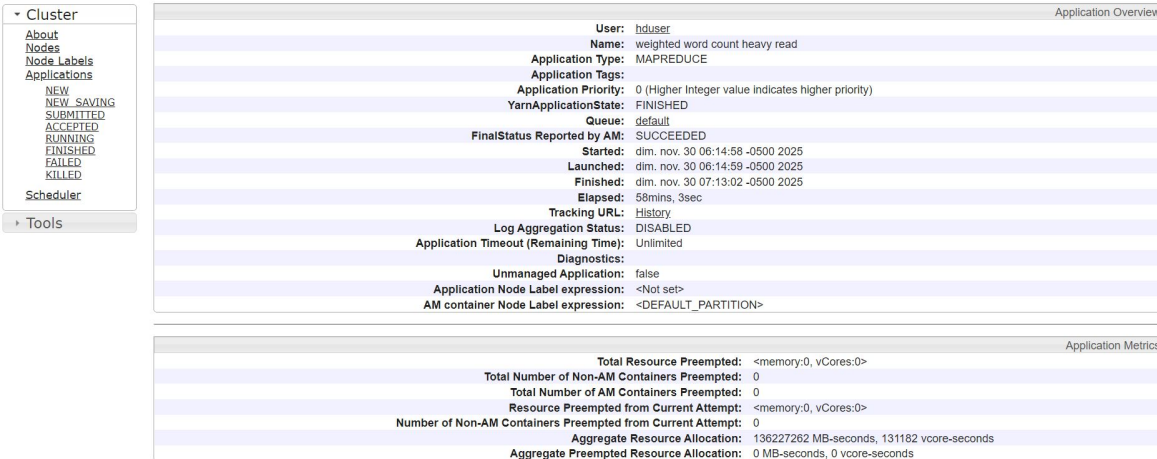


Figure 22: YARN application summary for the SBR WeightedWordCount job.

From the HDFS report, SBR increases space usage compared to the Default layout, but remains more conservative than ERMS. The NameNode reports a DFS used space of about 2.16 TB, which corresponds to roughly 79.6% of the available HDFS capacity. The per node view in Figure 23 shows that each DataNode stores between 210 GB and 218 GB, that is about 71% to 74% of the 294 GB local capacity. The distribution is very even across the ten DataNodes and the report lists no missing blocks and no under replicated blocks, which indicates that SBR preserves a healthy and well balanced cluster state while raising the replication factor for popular files.

Node	Http Address	Last contact	Last Block Report	Used	Non DFS Used	Capacity	Blocks	Block pool used	Version
✓/default-rack/vmhd-dn02:9866 (172.23.10.152:9866)	http://vmhd-dn02:9866	2s	253m	214.2 GB	64.35 MB	294.29 GB	3440	214.2 GB (72.79%)	3.3.6
✓/default-rack/vmhd-dn05:9866 (172.23.10.155:9866)	http://vmhd-dn05:9866	0s	157m	213.2 GB	64.14 MB	294.29 GB	3436	213.2 GB (72.45%)	3.3.6
✓/default-rack/vmhd-dn09:9866 (172.23.10.159:9866)	http://vmhd-dn09:9866	0s	47m	212.29 GB	64.17 MB	294.29 GB	3418	212.29 GB (72.14%)	3.3.6
✓/default-rack/vmhd-dn01:9866 (172.23.10.151:9866)	http://vmhd-dn01:9866	1s	144m	210.9 GB	4.65 GB	294.29 GB	3397	210.9 GB (71.66%)	3.3.6
✓/default-rack/vmhd-dn06:9866 (172.23.10.156:9866)	http://vmhd-dn06:9866	1s	287m	218.42 GB	64.25 MB	294.29 GB	3510	218.42 GB (74.22%)	3.3.6
✓/default-rack/vmhd-dn10:9866 (172.23.10.160:9866)	http://vmhd-dn10:9866	2s	131m	214.68 GB	64.19 MB	294.29 GB	3451	214.68 GB (72.95%)	3.3.6
✓/default-rack/vmhd-dn04:9866 (172.23.10.154:9866)	http://vmhd-dn04:9866	2s	349m	213.77 GB	64.23 MB	294.29 GB	3445	213.77 GB (72.64%)	3.3.6
✓/default-rack/vmhd-dn08:9866 (172.23.10.158:9866)	http://vmhd-dn08:9866	2s	332m	214.9 GB	64.15 MB	294.29 GB	3452	214.9 GB (73.02%)	3.3.6
✓/default-rack/vmhd-dn07:9866 (172.23.10.157:9866)	http://vmhd-dn07:9866	2s	84m	210.4 GB	5.14 GB	294.29 GB	3393	210.4 GB (71.49%)	3.3.6
✓/default-rack/vmhd-dn03:9866 (172.23.10.153:9866)	http://vmhd-dn03:9866	1s	135m	214.02 GB	64.19 MB	294.29 GB	3447	214.02 GB (72.72%)	3.3.6

Figure 23: Per node HDFS usage after the SBR replication plan has been applied.

The `hdfs fsck` output on `/exp/day2/data` confirms that SBR actively uses a mix of replication factors 2, 3, and 4 on the 10K most frequently accessed files. Files with modest demand are stored with replication factor 2, files with medium demand use factor 3, and the hottest files receive factor 4. All blocks are reported as OK and every replica is placed on distinct DataNodes, which shows how SBR concentrates additional replicas on high demand data while reducing the replication factor of less

popular items, without sacrificing fault tolerance.

The Ganglia overview in Figure 24 shows the typical shape of the SBR run. After an initial ramp up, the cluster enters a long phase where all nodes participate in the job, then activity drops sharply when the last containers finish. The aggregated network throughput curves in Figure 26 reveal a network intensive phase during roughly the first 20 to 25 minutes. Incoming traffic reaches between 2.4 and 2.8 GB/s at cluster level, while outgoing traffic peaks around 2.0 to 2.4 GB/s. After this initial burst, the throughput settles to a sustained plateau close to 0.7–1.0 GB/s until the end of the job. The packet rate in Figure 27 follows the same pattern, with aggregated incoming traffic up to about 1.6–1.7 million packets per second and outgoing traffic close to 60–70 thousand packets per second before both drop to a moderate steady state. This behavior is consistent with SBR creating and relocating replicas for the most popular files early in the run, then benefiting from improved locality during the long tail of repeated reads.

CPU and memory metrics confirm that SBR stresses the I/O subsystem more than compute. The aggregated user and system CPU graphs in Figure 28 show moderate utilization on the DataNodes: system time dominates user time, with short bursts that approach the full aggregate capacity when tasks start, but the overall average remains well below saturation. The cluster overview in Figure 24 reports an average CPU utilization around 41% over the measurement window, which leaves comfortable headroom on all nodes. The per node memory dashboards in Figure 25 indicate that each DataNode keeps several gigabytes of free memory and a large buffer cache throughout the run, with no sign of swapping. Overall, the selected SBR run shows moderate HDFS space growth together with sustained I/O activity and stable CPU and memory usage. The relative completion-time comparison across policies is based on the five-run statistics reported in the main manuscript.

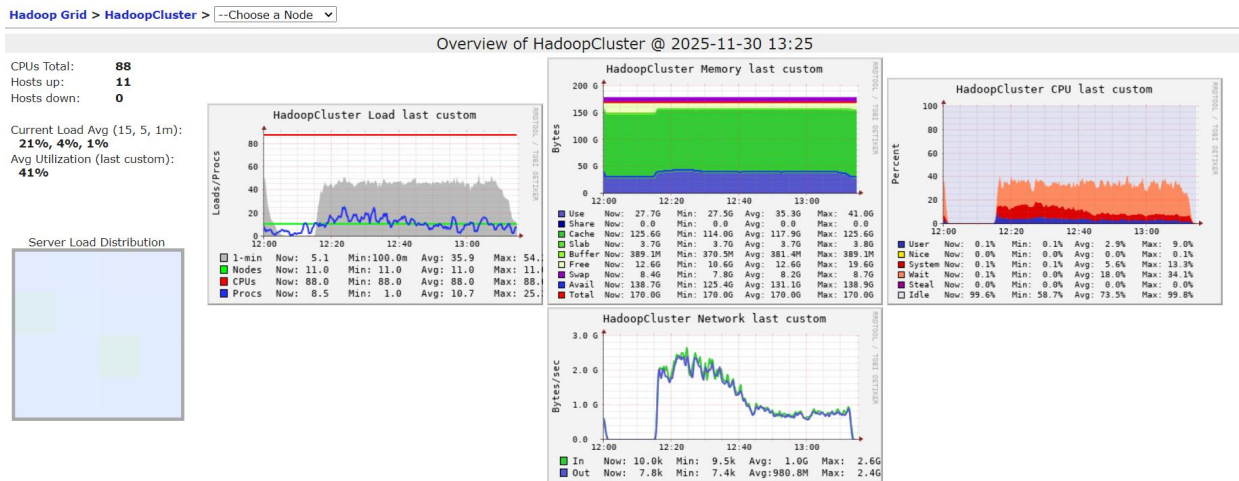


Figure 24: Ganglia overview of the Hadoop cluster during the SBR `WeightedWordCount` run.



Figure 25: Per node memory usage during the SBR run.

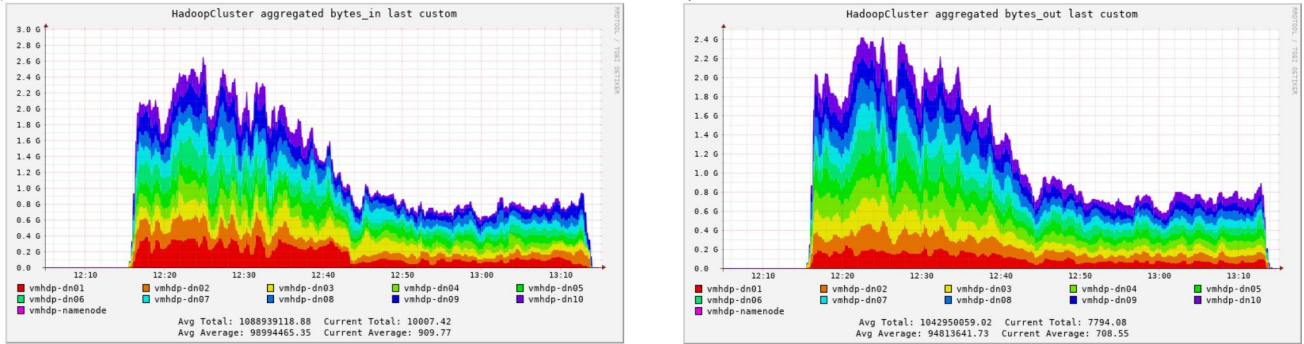


Figure 26: Aggregated incoming (left) and outgoing (right) network throughput for the SBR run.

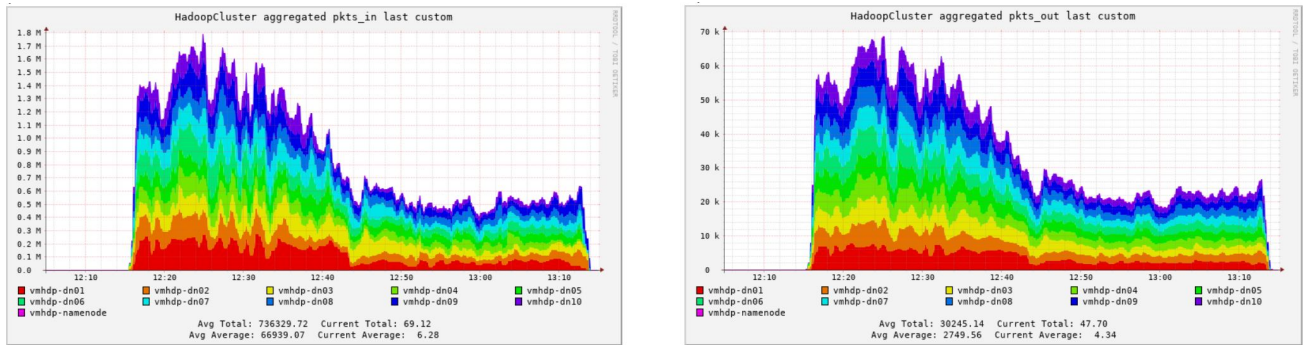


Figure 27: Aggregated incoming and outgoing packets per second for the SBR run.

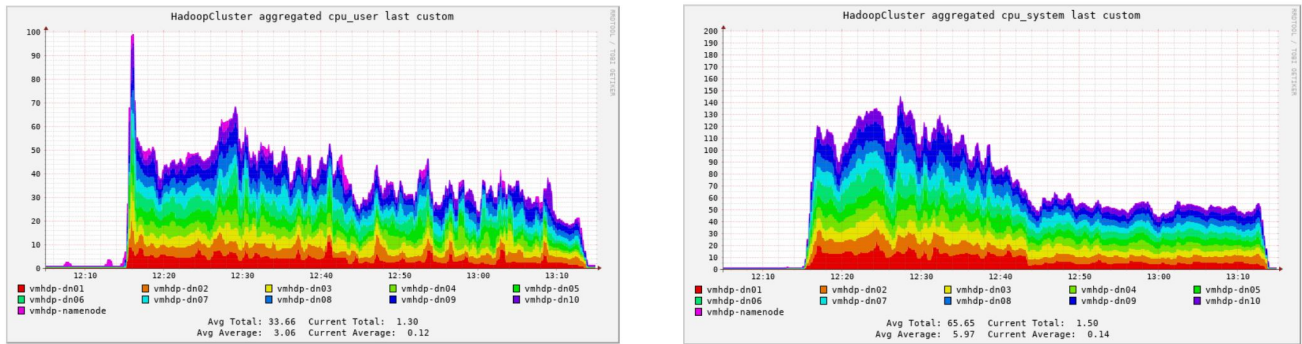


Figure 28: Per node user and system CPU utilization during the SBR run.

CLEAR:

Under the CLEAR layout, the heavy `WeightedWordCount` job keeps exactly the same logical workload as in the other policies. The YARN history for application `_0045` reports an elapsed time of about 50 min 35 s for the selected run shown in Figure 29. This screenshot documents the best observed CLEAR execution used for qualitative inspection in this appendix. The quantitative comparison with the other policies is based on the five-run statistics reported in the main manuscript.

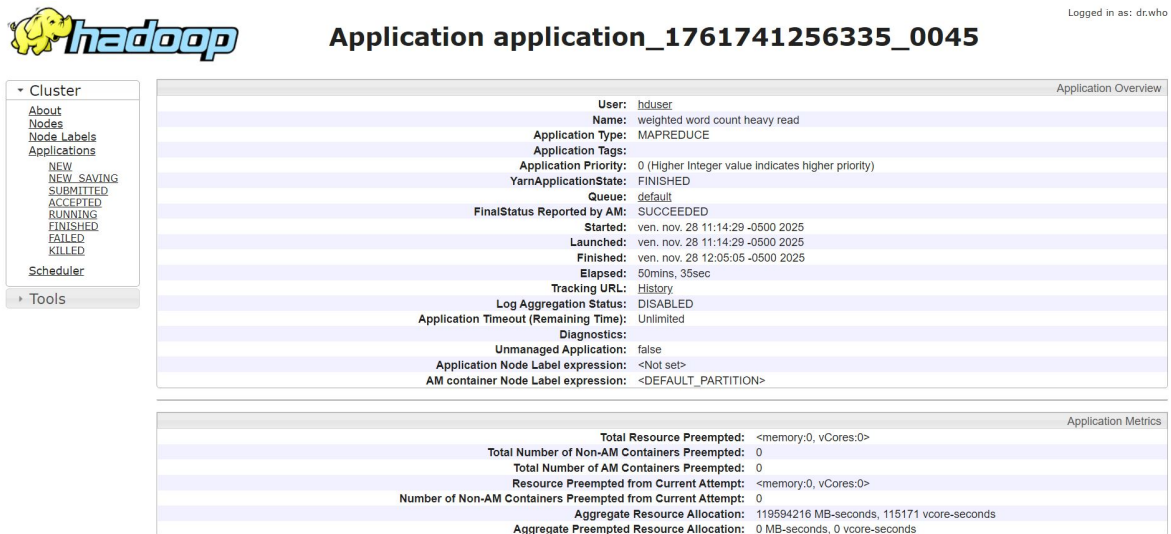


Figure 29: YARN application summary for the CLEAR `WeightedWordCount` job.

The HDFS counters confirm that the job issues 6.7 TB of HDFS reads in total, and the custom `WeightedHeavyRead` counter reports a very similar number of bytes, which shows that the logical workload is unchanged. In addition, the counter `HDFS: Number of bytes read erasure-coded` reaches about 280 GB, which means that a visible fraction of the requests is served directly from erasure coded stripes. Among the four layouts, CLEAR is the only one where some of the 10K most frequently accessed files are actually stored in the erasure coded layout instead of purely replicated form.

Node	Http Address	Last contact	Last Block Report	Used	Non DFS Used	Capacity	Blocks	Block pool used	Version
✓ /default-rack/vmhdn-dn02:9866 (172.23.10.152:9866)	http://vmhdn-dn02:9866	1s	59m	224.03 GB	64.12 MB	294.29 GB		4254 224.03 GB (76.13%)	3.3.6
✓ /default-rack/vmhdn-dn05:9866 (172.23.10.155:9866)	http://vmhdn-dn05:9866	1s	323m	233.14 GB	64.12 MB	294.29 GB		4433 233.14 GB (79.22%)	3.3.6
✓ /default-rack/vmhdn-dn09:9866 (172.23.10.159:9866)	http://vmhdn-dn09:9866	2s	214m	228.63 GB	64.12 MB	294.29 GB		4355 228.63 GB (77.69%)	3.3.6
✓ /default-rack/vmhdn-dn01:9866 (172.23.10.151:9866)	http://vmhdn-dn01:9866	1s	310m	226.35 GB	4.65 GB	294.29 GB		4310 226.35 GB (76.91%)	3.3.6
✓ /default-rack/vmhdn-dn06:9866 (172.23.10.156:9866)	http://vmhdn-dn06:9866	1s	93m	221.61 GB	64.12 MB	294.29 GB		4239 221.61 GB (75.3%)	3.3.6
✓ /default-rack/vmhdn-dn10:9866 (172.23.10.160:9866)	http://vmhdn-dn10:9866	0s	298m	218.03 GB	64.14 MB	294.29 GB		4173 218.03 GB (74.09%)	3.3.6
✓ /default-rack/vmhdn-dn04:9866 (172.23.10.154:9866)	http://vmhdn-dn04:9866	2s	155m	224.89 GB	64.12 MB	294.29 GB		4270 224.89 GB (76.42%)	3.3.6
✓ /default-rack/vmhdn-dn08:9866 (172.23.10.158:9866)	http://vmhdn-dn08:9866	1s	139m	214.74 GB	64.13 MB	294.29 GB		4121 214.74 GB (72.97%)	3.3.6
✓ /default-rack/vmhdn-dn07:9866 (172.23.10.157:9866)	http://vmhdn-dn07:9866	1s	251m	224.52 GB	5.1 GB	294.29 GB		4287 224.52 GB (76.29%)	3.3.6
✓ /default-rack/vmhdn-dn03:9866 (172.23.10.153:9866)	http://vmhdn-dn03:9866	0s	302m	226.73 GB	64.13 MB	294.29 GB		4334 226.73 GB (77.04%)	3.3.6

Figure 30: Per node HDFS usage after the CLEAR replication plan has been applied.

The consistency report on `/exp/day2/data` shows how CLEAR combines selective replication with erasure coding. The `fsck` output lists 8 622 files and 8 622 blocks, all reported as OK, with no missing, under replicated, or corrupted blocks. Every file is stored as a single 64 MB block whose replication factor varies between 1 and 5. Most blocks use replication levels between 2 and 5, with a distribution of 581 blocks at RF 2, 3 275 at RF 3, 1 543 at RF 4, and 3 222 at RF 5, plus a single file at RF 1. This corresponds to an average replication factor of about 3.86 on this directory, which is higher than the Default layout but obtained in a targeted way, only for the most active part of the workload. Colder files that do not belong to the 10K hot set are placed in a separate erasure coded layout and are not replicated at high factors.

The global HDFS report taken after applying the CLEAR plan indicates that the cluster runs at a comfortable utilization level. The NameNode reports 2.19 TB of DFS used space, which corresponds to about 80.6% of the available HDFS capacity. At DataNode level (Figure 30), used space ranges from roughly 221 GB to 233 GB per node, that is about 75% to 79% of the 294 GB local capacity. The difference between the lightest and the heaviest node stays within a few percentage points, which confirms that CLEAR preserves a well balanced HDFS layout. The HDFS summary also reports zero missing blocks, zero under replicated blocks, and no degraded erasure coded block groups, so the more aggressive mix of replication factors and EC does not weaken fault tolerance.

The Ganglia overview in Figure 31 reflects this behavior at runtime. After a short ramp up, the cluster enters an intense phase where all nodes participate in the job, followed by a sharper drop in activity once the last containers finish. The aggregated throughput curves in Figure 33 show a pronounced network intensive period between roughly 17:20 and 17:55. Incoming traffic climbs to about 2.4–2.8 GB/s at cluster level, while outgoing traffic peaks around 2.0–2.4 GB/s. Compared with SBR and ERMS, the amplitude of the peaks is similar, but the plateau is shorter, which matches the reduced completion time. After the main burst, the throughput falls quickly to near zero, since the job terminates earlier and there is no long tail of remote reads.

The packet level metrics in Figure 34 follow the same pattern. Aggregated incoming traffic reaches roughly 1.6–1.7 million packets per second at the height of the job, while outgoing traffic stays around 50–60 thousand packets per second. Once the replication and read traffic stabilize, both curves decrease and the cluster becomes almost idle again. CPU metrics confirm that CLEAR stresses I/O more than compute. The per node user and system CPU graphs in Figure 35 show moderate utilization: there are

short bursts when tasks start and when replicas are created or accessed, but the overall level remains well below saturation, with a large fraction of idle time even during the core of the run. The per node memory dashboards in Figure 32 indicate that all DataNodes keep several gigabytes of free memory and a large buffer cache, with no visible swapping. Overall, the selected CLEAR run shows a balanced cluster state, stable memory behavior, and sustained I/O activity while using both selective replication and erasure coding. The performance conclusion in the main manuscript is based on the five repeated executions rather than on this single displayed trace.

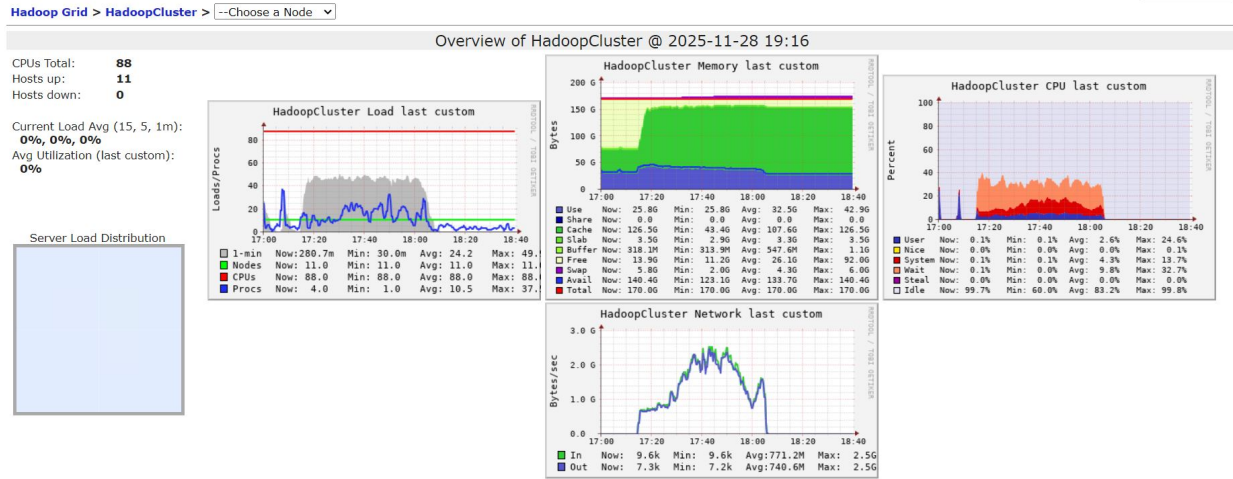


Figure 31: Ganglia overview of the Hadoop cluster during the CLEAR WeightedWordCount run.



Figure 32: Per node memory usage during the CLEAR run.

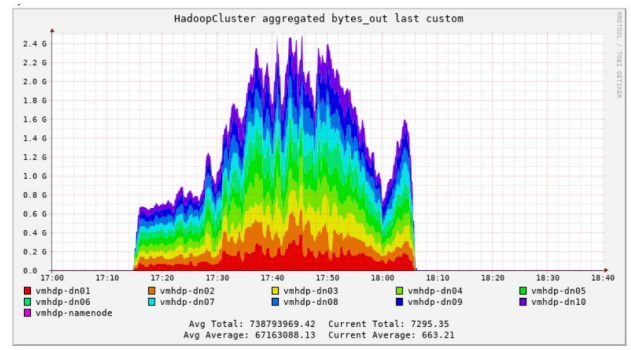
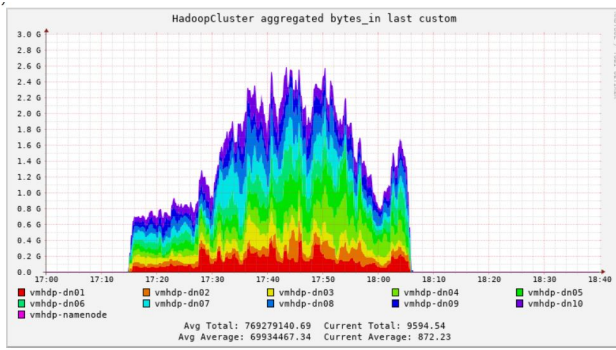


Figure 33: Aggregated incoming (left) and outgoing (right) network throughput for the CLEAR run.

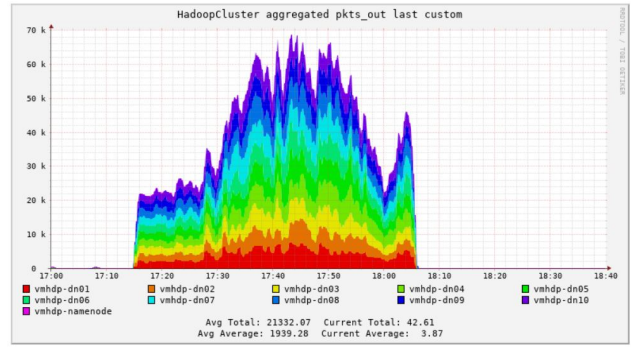
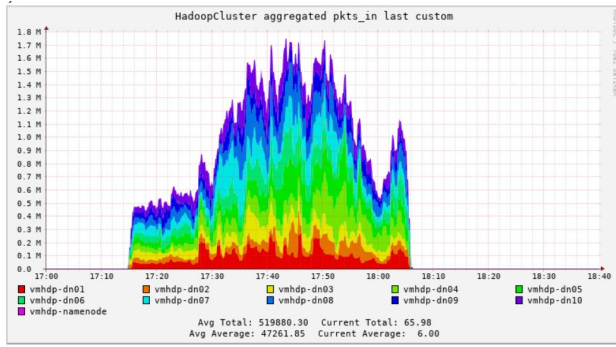


Figure 34: Aggregated incoming and outgoing packets per second for the CLEAR run.

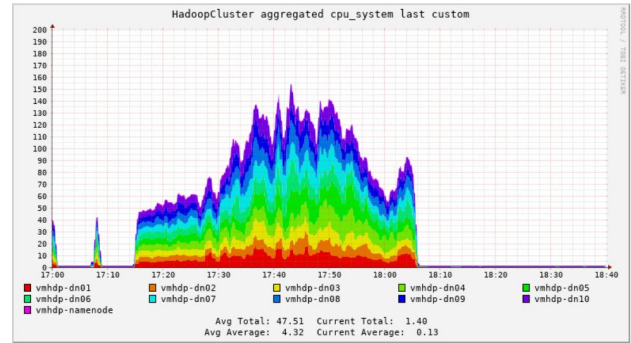
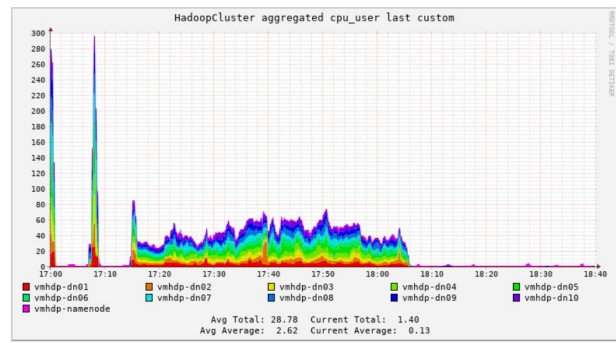


Figure 35: Per node user and system CPU utilization during the CLEAR run.